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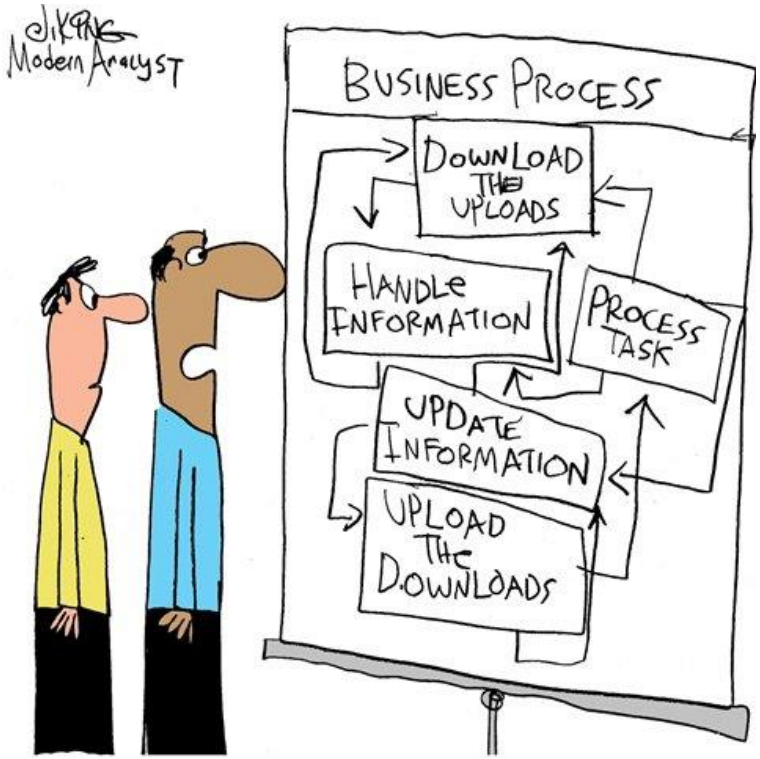
Dilshan Jayatilake

UFCFTR-30-3
Distributed & Enterprise Software Development

Introduction to Business Process Modelling

Session Plan

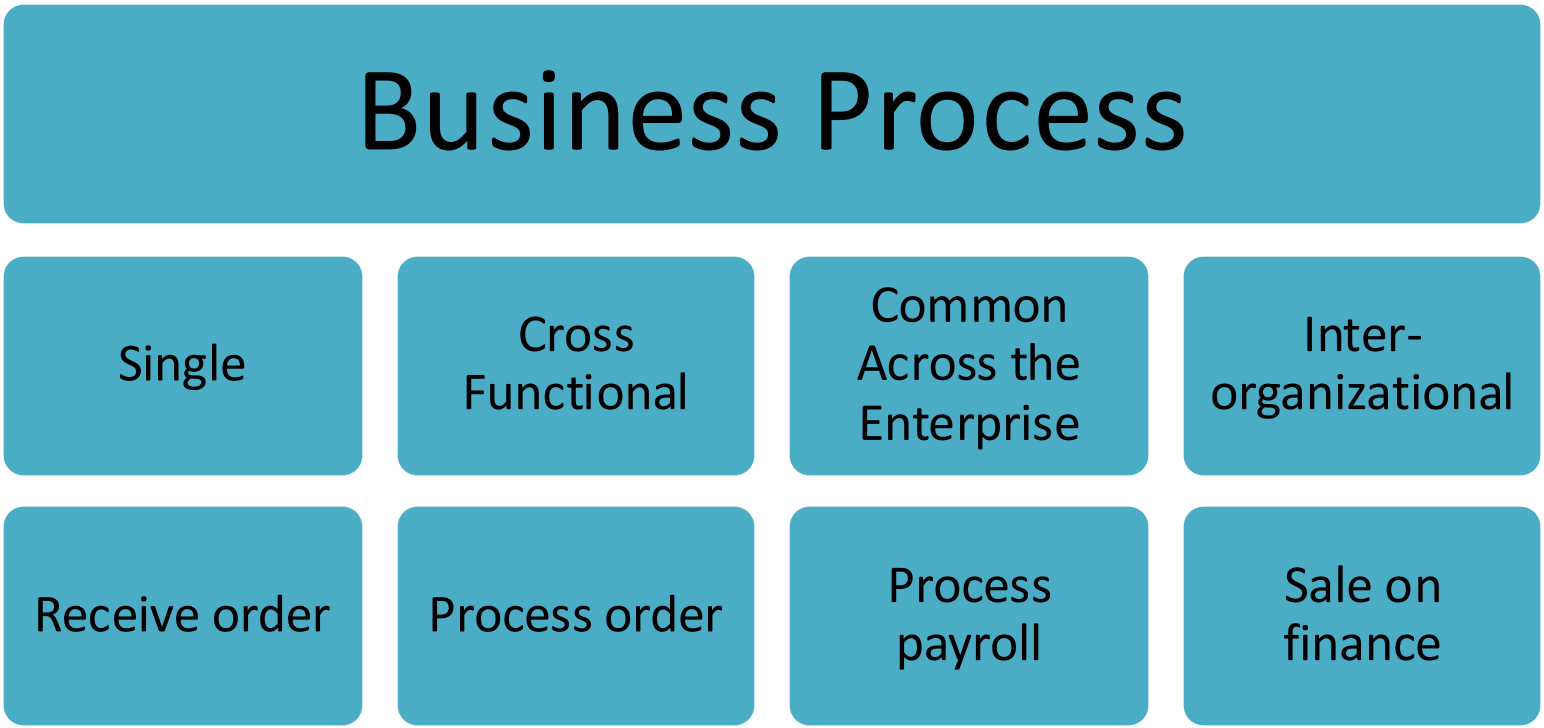
- Introduction to Business Process & Business Process Management
- Business Process Modelling (BPM)
  - What is modelling?
  - What is BPM and why?
- Business Process Model and Notation (BPMN)
  - Introduction and background
  - BPMN Vs UML
- Business Process Modelling Tools (Camunda)
  - What is Camunda Modeler
  - What is Camunda (Over view)



"Charlie, did you really get these details from the customer?"

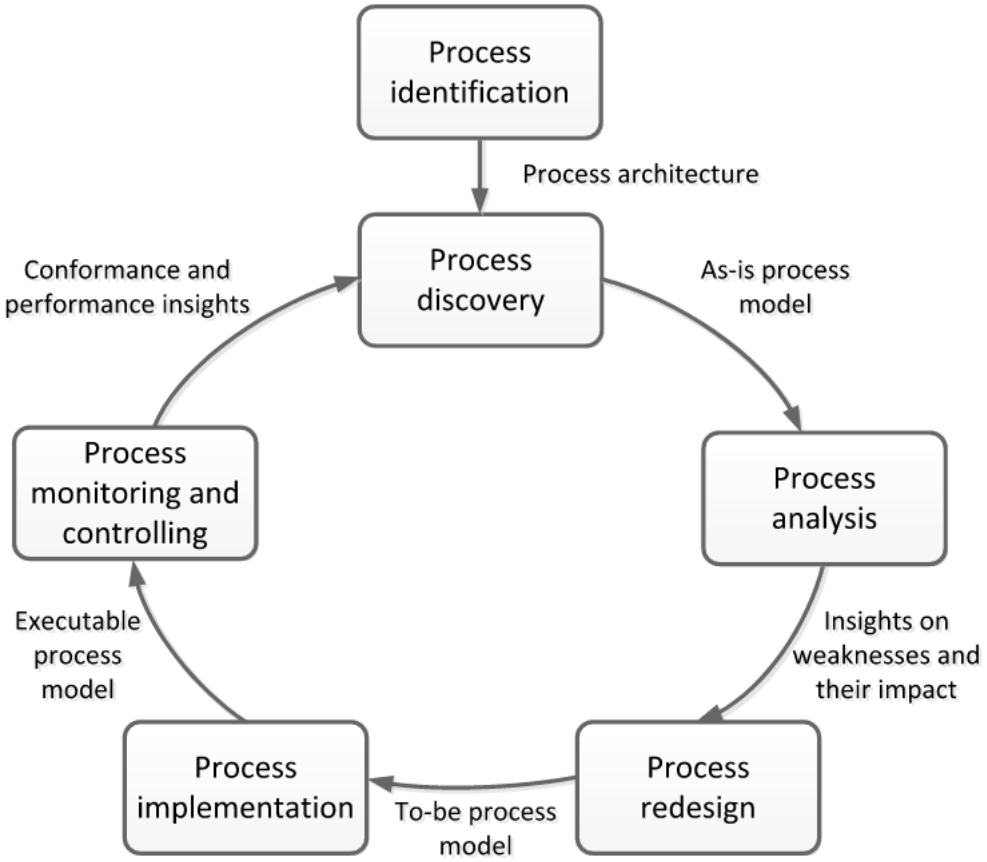
Business Process

"Business Process is a ***collection*** *of* ***inter-related events, activities and decision points*** *that involve a number of* ***actors and objects****, and that collectively lead to an outcome that is of value to at* ***least one customer****."* (Dumas *et al.*, 2013)



Business Process Management

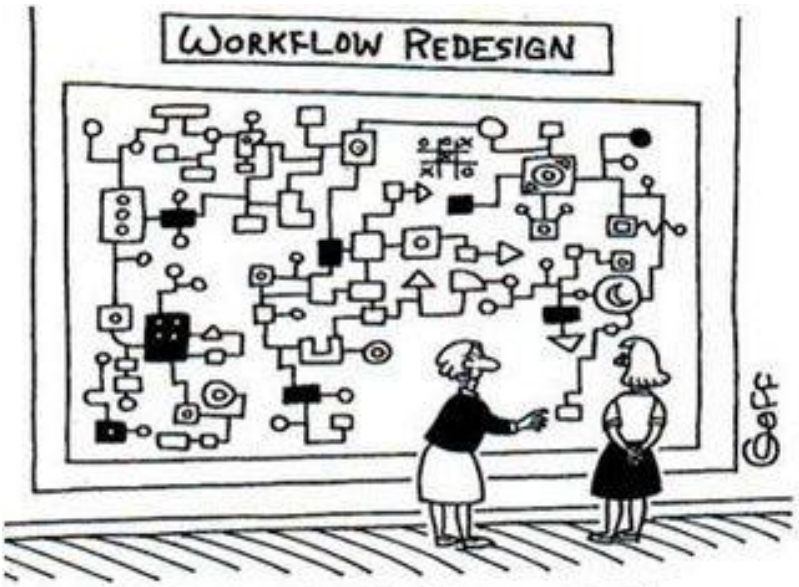
Business Process Management is defined as: "*body of methods, techniques and tools to discover, analyse, redesign, execute and monitor business processes*" (Dumas *et al.*, 2013)



(Dumas *et al.*, 2013)

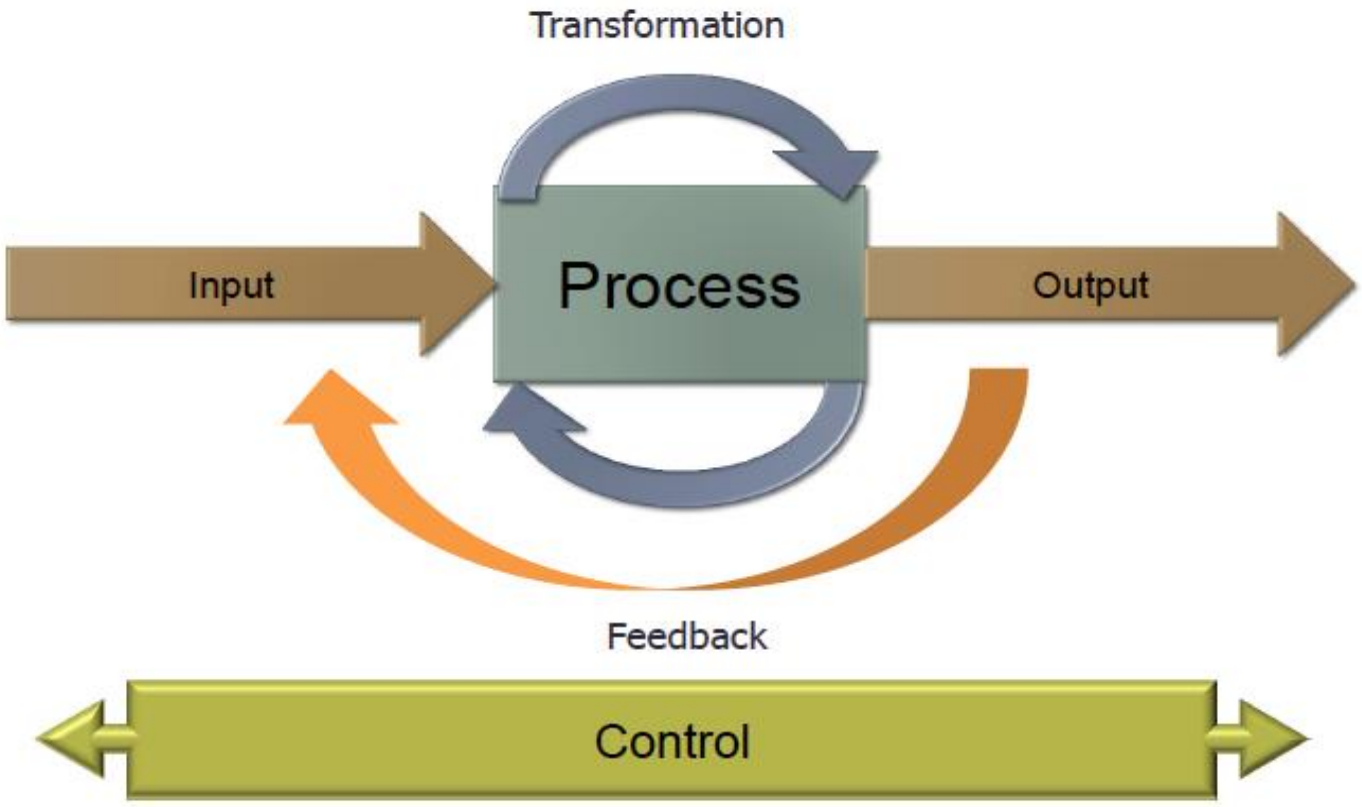
Challenges in Business Process Management

1. Resistance to change and people management
2. Lack of **consistent understanding** of the reason for change
3. Risk that the final solution does not **meet business expectations**
4. Lack of **consistent processes**
   - re-documenting the same processes that have already been documented
5. Absence of a **common language**
   - Culture on change for sustaining the new ways of working



"And this is where our ED workflow redesign team went insane."

Business Process & Business Process Management

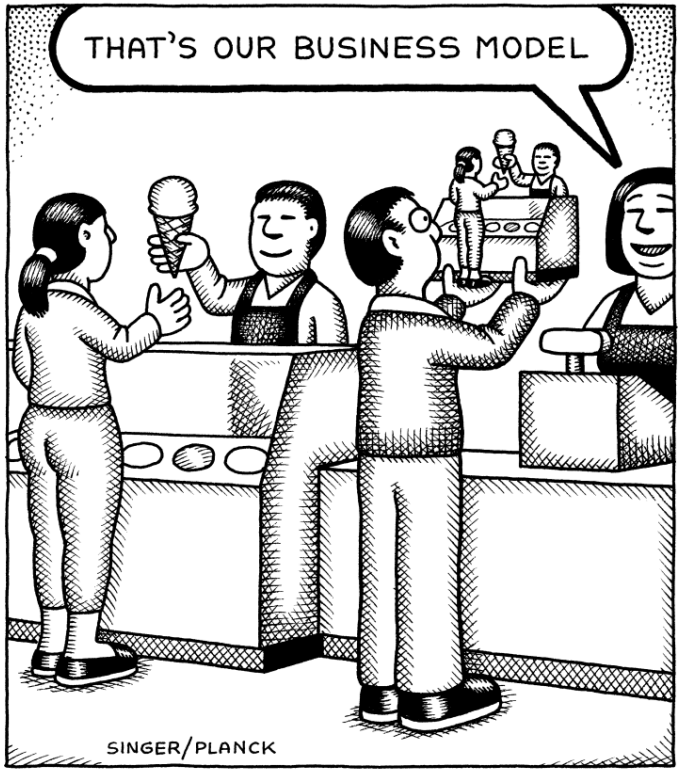


What is modelling?

- Mapping of reality to serve a specific purpose
- Expressed in
  - Symbols, Types, and Words
- Represents
  - Entities, Processes, Attributes and their Relations
- Facilitates
  - Description
  - Understanding
  - Complexity
  - Context
- Focus on
  - specific aspects of reality and degrade or ignore the rest
  - certain time and a certain purpose (Pragmatics)

Rule Of Thumb for Modelling?

1. Correctness
   - Capture attributes
2. Relevance
   - Level of abstraction
3. Economic Efficiency
   - Don’t forget the objective
4. Clarity
   - Facilitate understanding
5. Comparability
   - Easy to relate on common aspects
6. Systematic Design
   - Common interfaces



Business Process Modelling Types

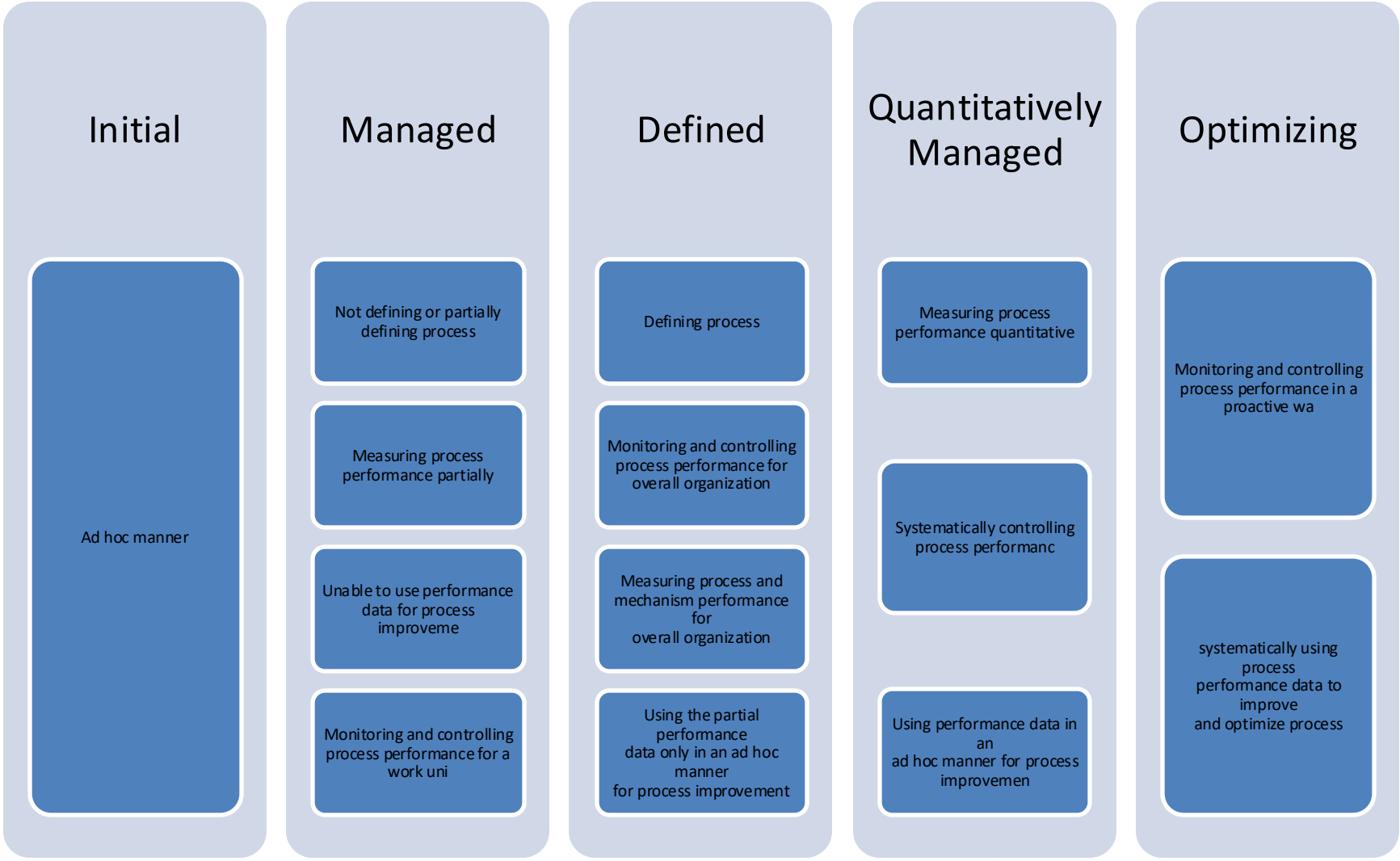
Dynamic Model Types

- Process relevant information
- Chronological / Time dependent
- Event Driven Process Chains
- Value Added Chain Diagrams

Static Model Types

- No Time Dependency
- Organizational structures
- Information Carriers
- Relationships between business object

BPM - Maturity Level



(Lee, Lee and Kang, 2007)

BPM in action

- **Modelling Language**
  - UML (Unified Modelling Language)
  - BPMN (Business Process Modelling Notation)
- **Modelling Framework**
  - Business Process Lifecycle
  - Specificity
  - Generalisability
- **Enterprise Architecture**
  - Zachman Framework
  - CIMOSA Framework for Modelling
  - GERA Modelling Framework
  - i*

Business Process Model and Notation (BPMN)

- BPMN
  - A **notation standard** for modelling business processes and services
  - Originally defined by the **Business Process Management Initiative** (BPMI) in 2004
  - Adopted by **Object Management Group** (OMG) in 2006
- BPMN Goals
  - Notation easily conceivable by all users
  - Provide technical detail for implementation
  - Aid interoperability between Business Process Management tools





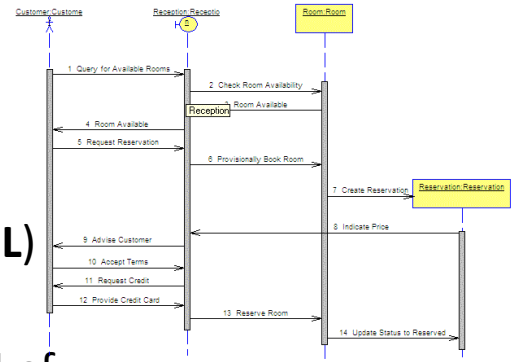
BPMN Diagram (Overview)

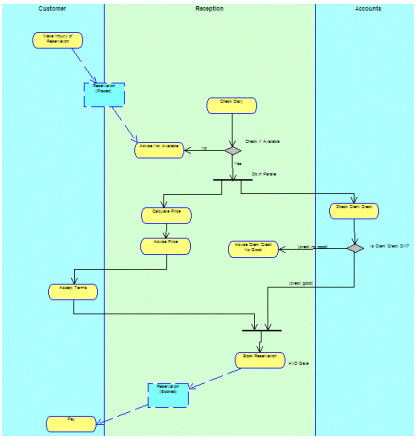
- A notation standard for all users
  - **Business analysts**
    - create drafts of the process models
  - **Technical developers**
    - implement the processes
  - **Business people**
    - that manage the processes



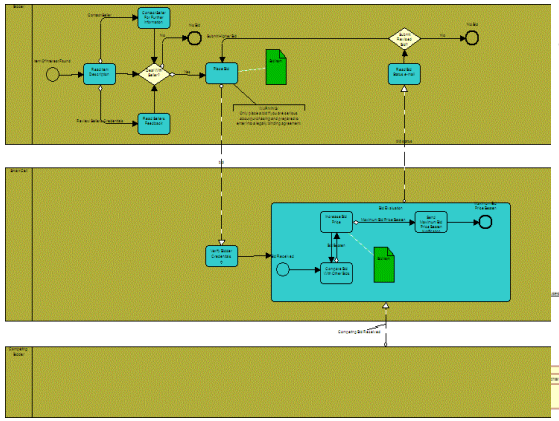
- Capabilities for
  - **Exception Handling**
    - Information exchange, Business rules, Legal framework
  - **Transactions**
    - Check points, activities, sequence and data
  - **Compensation**
    - Supportive information systems, digital artefacts
- **Interconnection with**
  - Execution languages like BPEL (Business Process Execution Language)

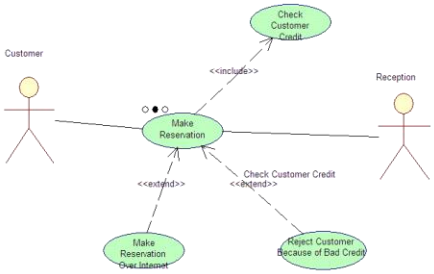
BPMN vs UML

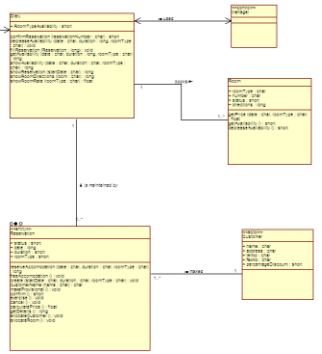




- **Unified Modelling Language (UML)**
  - Standardized, general-purpose modelling language in the field of software engineering
  - UML describes system behavior
- UML (Object Oriented) # BPMN (Process Oriented)
- BPMN and UML can co-exist
- BPMN **is not** UML Activity Diagram
  - Activity Diagrams
    - modeling algorithms
    - subset of BPMN
  - BPMN
    - superior for business process modelling
    - asynchronous message exchanges







BPMN Model Interchange Working Group (BPMN MIWG)

- Purpose
  - The purpose of the BPMN MIWG is to support, facilitate, and promote the interchange of BPMN Models
- Mandate
  - Facilitate and demonstrate BPMN Model Interchange amongst tools that implement the standard
  - Resolve vendors' misinterpretation of the specification that inhibit BPMN Model Interchange
  - Identify and report defects that break BPMN Model Interchange
  - Establish a BPMN Model Interchange demonstration infrastructure to support the above, including:
    - Demonstration Processes,
    - Guidelines, and
    - Validation tools.

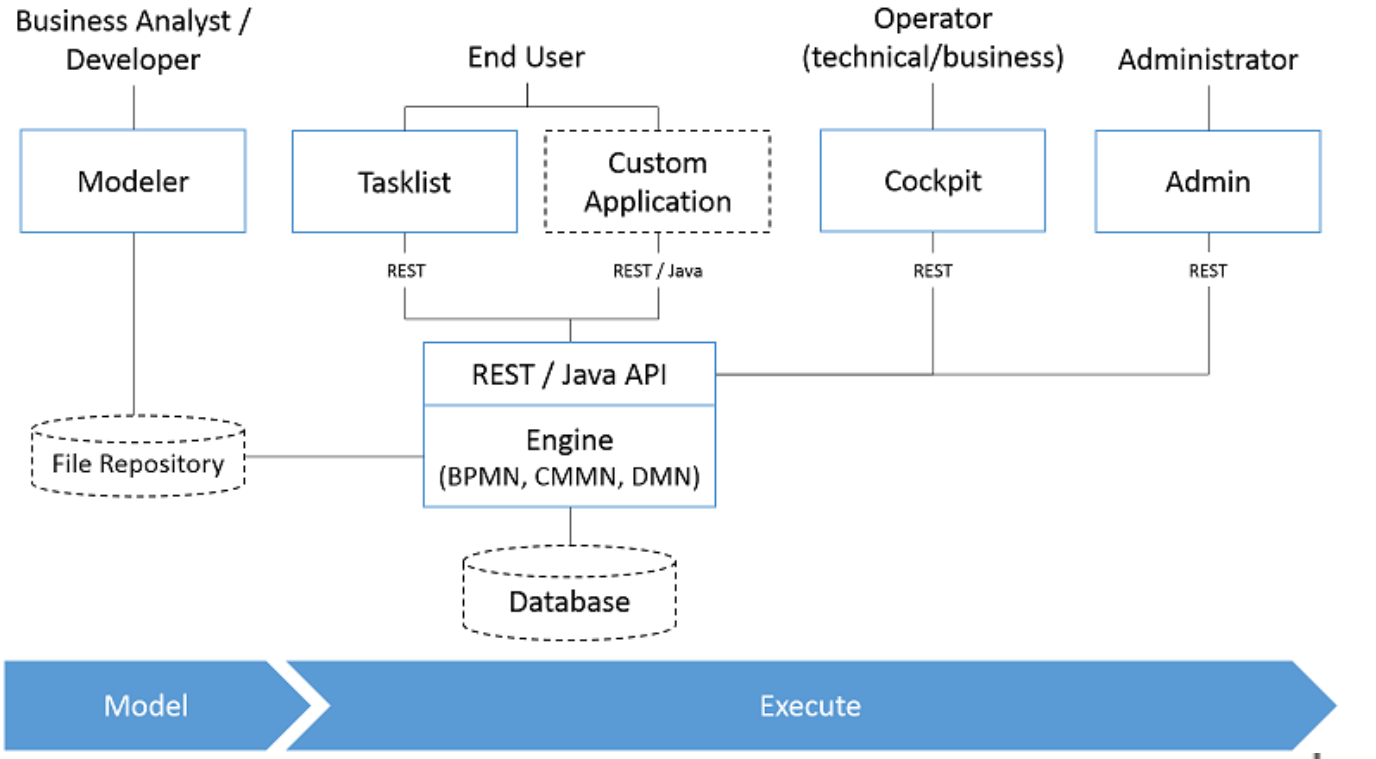


Camunda

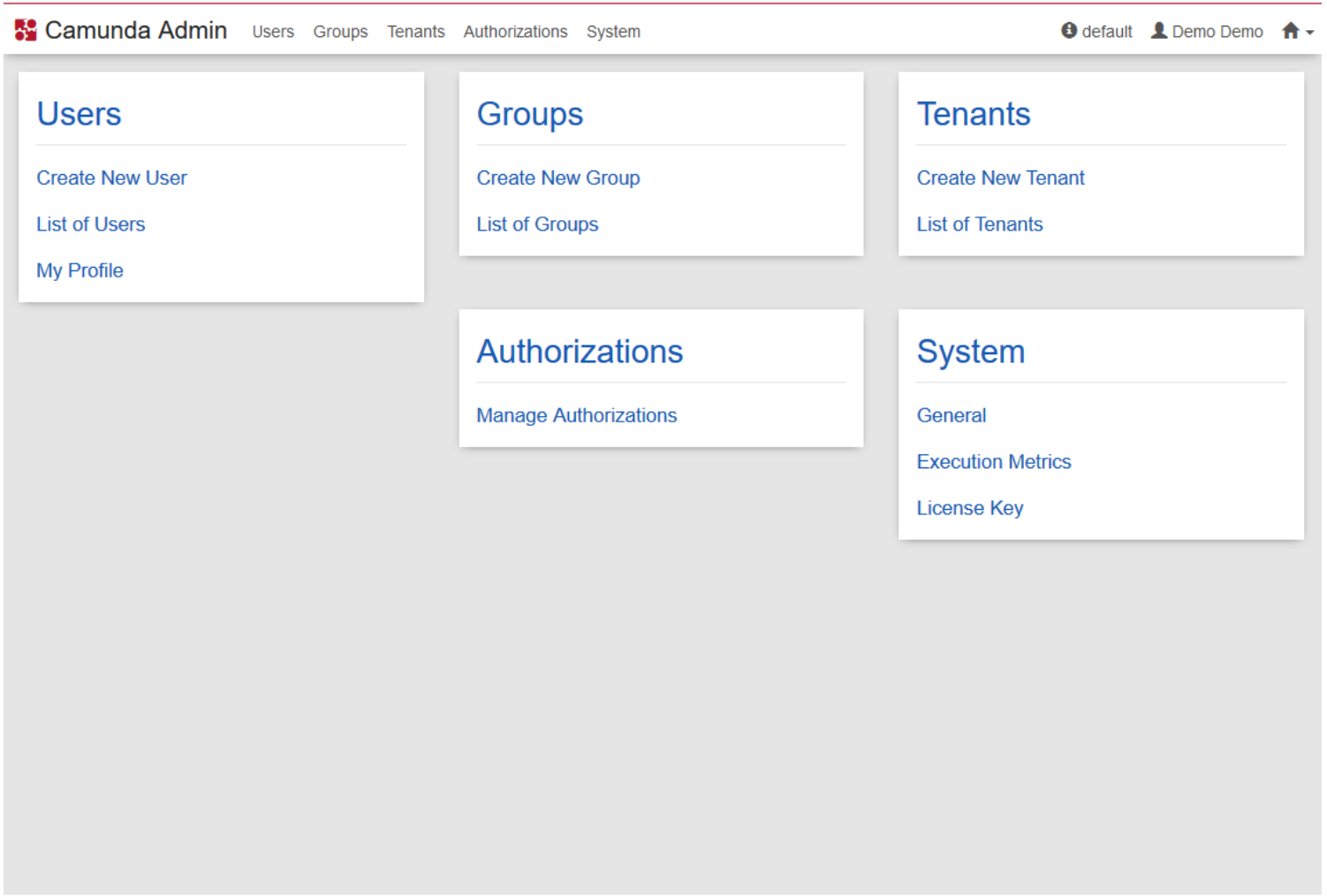


Camunda

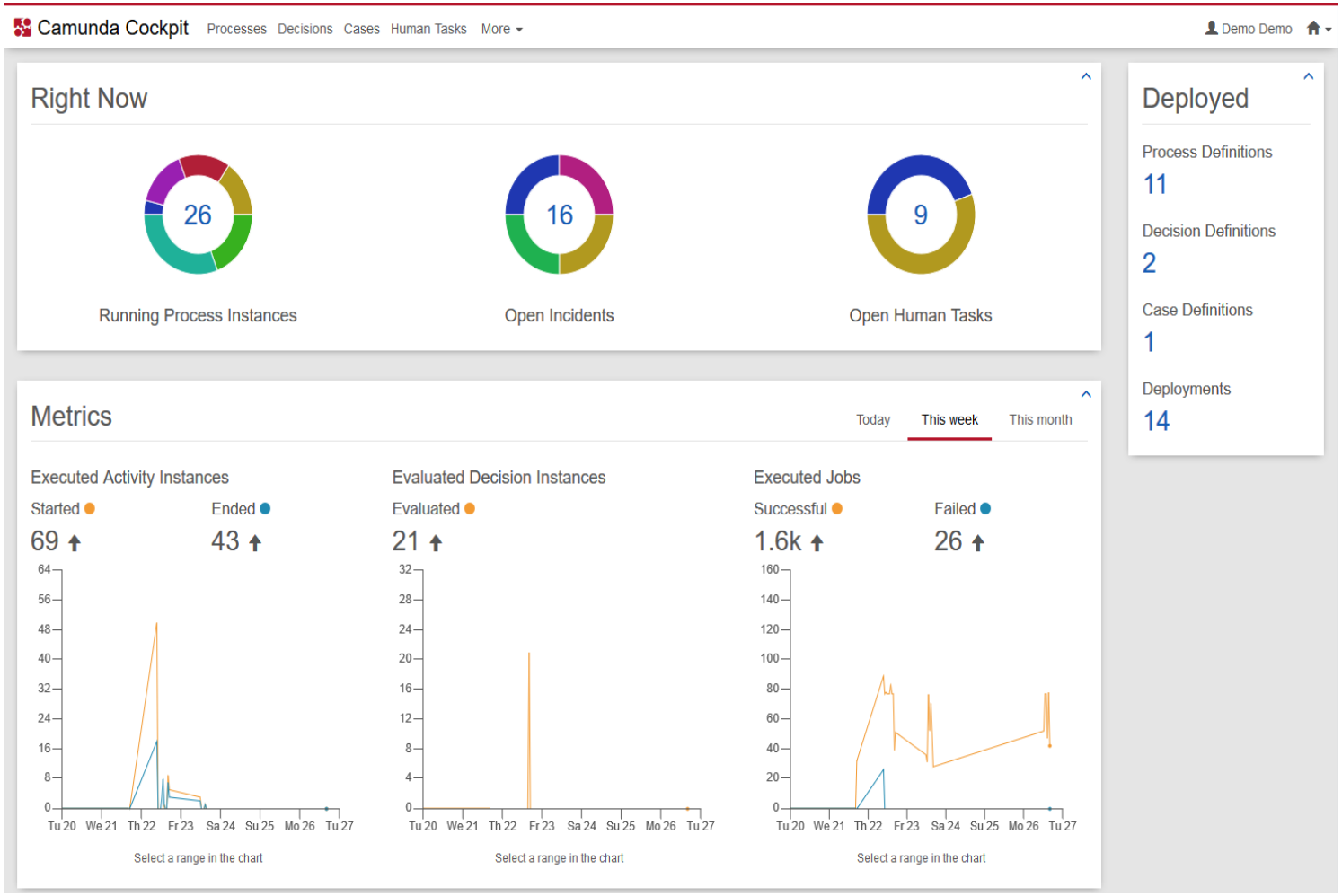
- Camunda (https://camunda.com/)
  - Java-based framework supporting BPMN for workflow and process automation



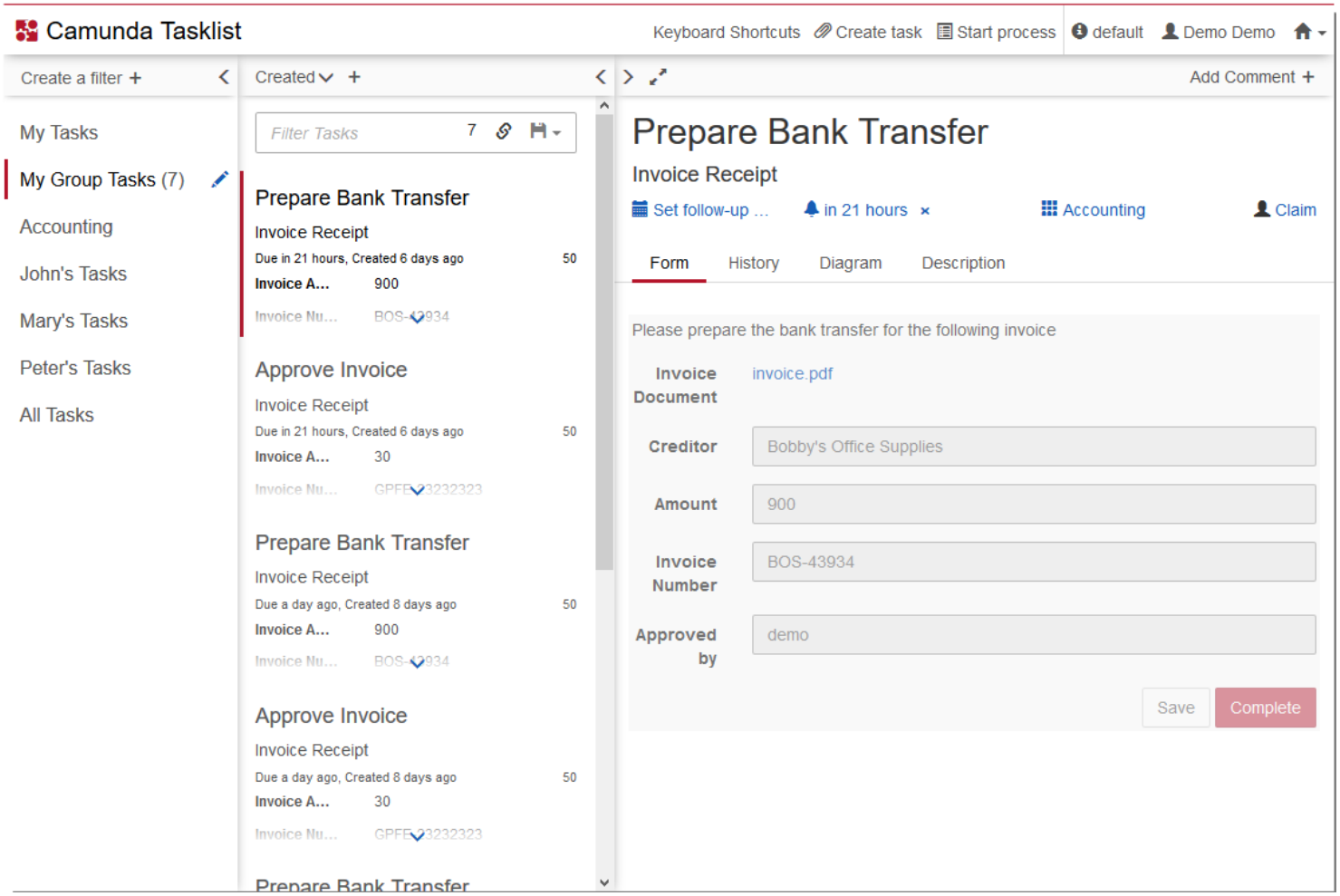
Camunda Admin



Camunda Cockpit



Camunda Tasklist



Any Questions?



Go to menti.com and use the code **7239 6346**

OR

https://www.menti.com/al7jed5ggrv5

References and Further Reading

- Hoggett, C. and Shah, P., 2013. Maximizing the value of Enterprise Process Modeling: Process governance in large scale transformation programs. *BPTrends*.
- Schuette, R. and Rotthowe, T., 1998. The guidelines of modeling–an approach to enhance the quality in information models. In *Conceptual Modeling–ER'98* (pp. 240-254). Springer, Berlin, Heidelberg.
- Zurich Financial Services Group Annual Report 2005 https://www.zurich.com/_/media/dbe/corporate/docs/financial-reports/archive/financial-report-2005.pdf?la=en
- Object Management Group, Business Process Model and Notation. http://www.bpmn.org/

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